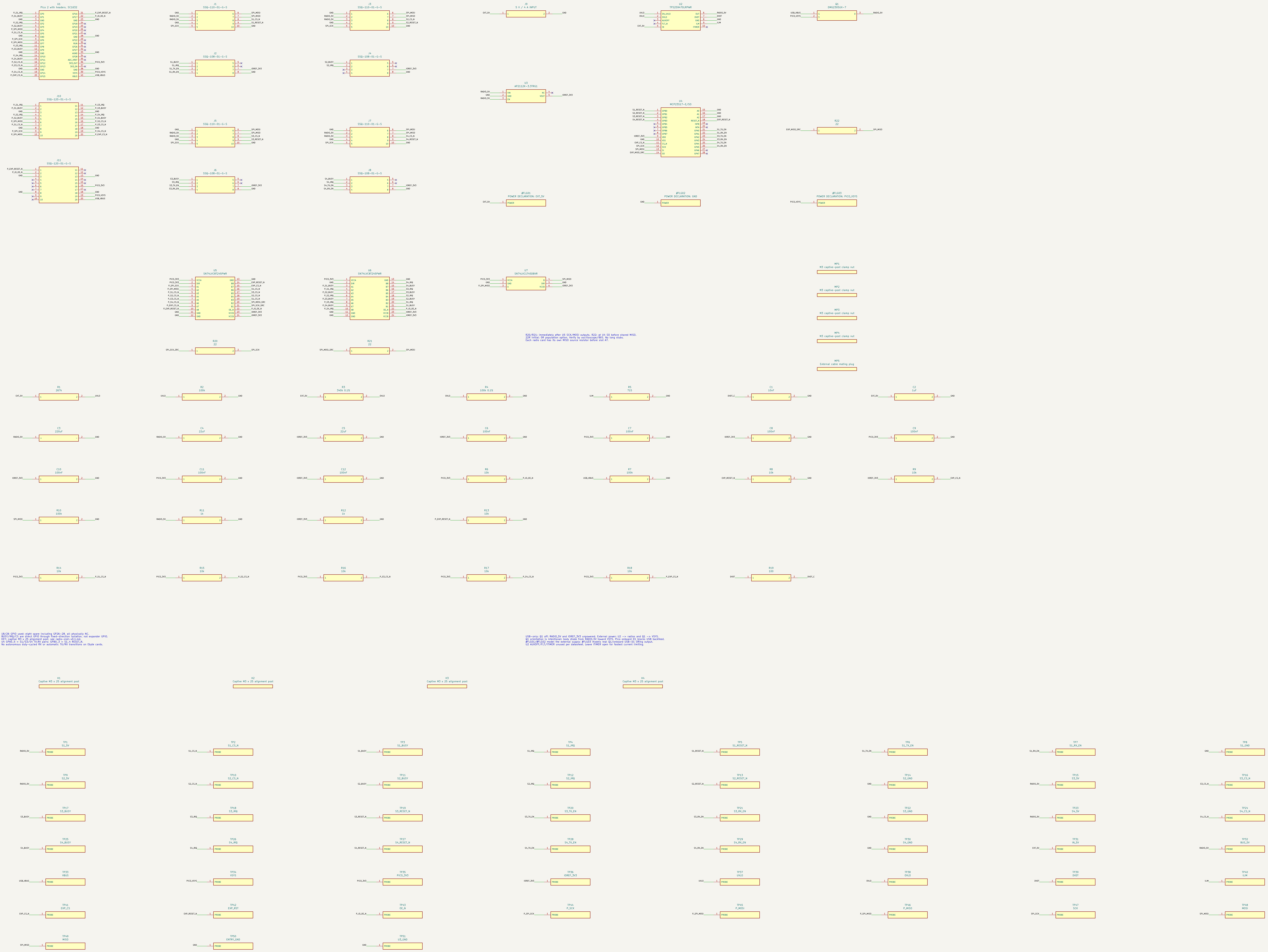
REV A LAYOUT REVIEW. NOT FOR FABRICATION.

Four-slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.


U1
Pico 2 with headers, SC1632
J1
SSQ-110-01-G-S
J9
5 V / 4 A INPUT
U2
TPS259470LRPWR
Q1
DMG2305UX-7
U3
AP2112K-3.3TRG1
U4
MCP23S17-E/SO
R22
22
#FLG01
POWER DECLARATION: EXT_5V
#FLG02
POWER DECLARATION: GND
#FLG03
POWER DECLARATION: PICO_VSYS
U5
SN74LVC8T245PWR
U6
SN74LVC8T245PWR
U7
SN74LVC1T45DBVR
R20
22
R21
22
MP1
M3 captive-post clamp nut
MP2
M3 captive-post clamp nut
MP3
M3 captive-post clamp nut
MP4
M3 captive-post clamp nut
MP5
External cable mating plug
R2
100k
R3
340k 0.1%
R4
100k 0.1%
R5
715
C1
10nF
C2
1uF
H1
Captive M3 x 25 alignment post
H2
Captive M3 x 25 alignment post
H4
Captive M3 x 25 alignment post

R20/R21: immediately after U5 SCK/MOSI outputs. R22: at U4 SO before shared MISO.
22R initial; 0R population option. Verify by oscilloscope/IBIS. No long stubs.
Each radio card has its own MISO source resistor before slot A7.

18/26 GPIO used; eight spare including GP26–28, all physically NC.
BUSY/IRQ/CS are direct GPIO through fixed–direction isolation, not expander GPIO.
KEY: captive M3 x 25 alignment post, see radio–slot–v0.1.md.
U4 GPA0..5 = S1/S3/S4 TX,RX pairs; GPB0..3 = S1..4 RESET_N.
No autonomous duty–cycled RX or automatic TX/RX transitions on Ebyte cards.

USB–only: Q1 off; RADIO_5V and IOREF_3V3 unpowered. External power: U2 –> radios and Q1 –> VSYS.
Q1 orientation is intentional: body diode from RADIO_5V toward VSYS. Pico onboard D1 blocks USB backfeed.
#FLG01/#FLG02 model the external supply; #FLG03 models real Q1/onboard USB–D1 ORing output.
U2 AUXOFF/FLT/ITIMER unused per datasheet. Leave ITIMER open for fastest current limiting.

SCHEMATIC REVIEW – NOT FOR FABRICATION

Sheet: /
File: carrier–rev–a.kicad_sch
Title: Four–slot carrier: protected external 5 V / 4 A, isolated GPIO
Size: A0 | Date: 2026–09–17 | **Rev: A–schematic**
KiCad E.D.A. 10.0.6 | Id: 1/1